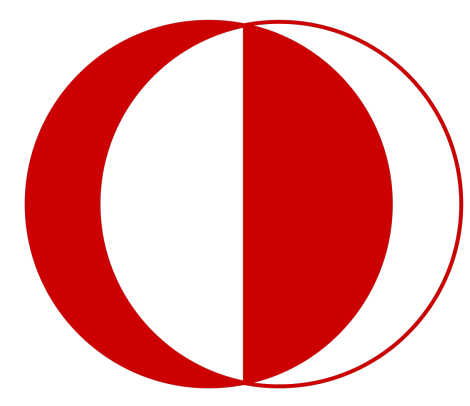




Automated Construction of Multi-Element MEAM Interatomic Potentials

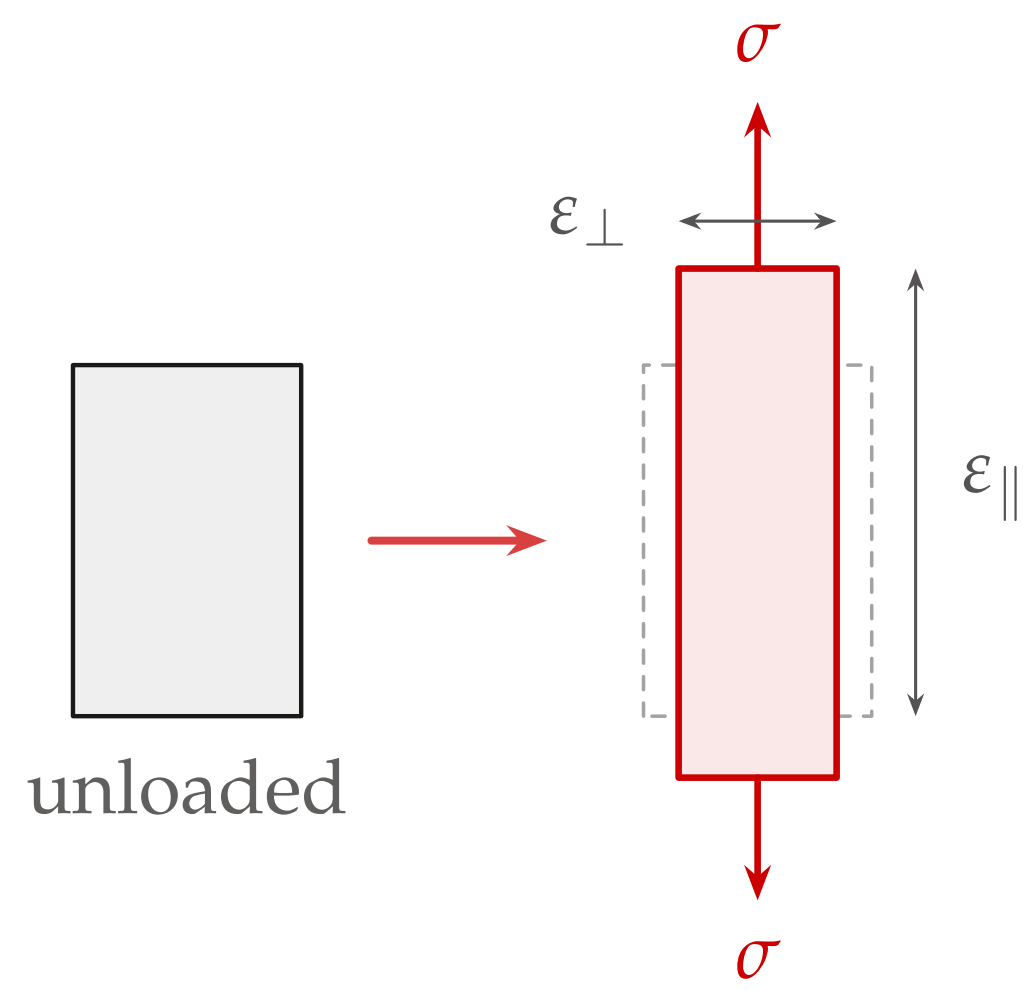
A DFT-informed neural-network optimization framework

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1 Motivation & background

Two numbers govern how a structural alloy carries load — and a neural network can predict both from composition alone.



Young's modulus $E = \sigma / \epsilon_{\parallel}$ — stiffness, the stress needed to stretch it (GPa).

Poisson's ratio $\nu = -\epsilon_{\perp} / \epsilon_{\parallel}$ — how much it thins sideways when pulled.

For a cubic crystal both reduce to just two stiffness constants — C_{11} (axial) and C_{12} (transverse):

$$E = \frac{(C_{11} - C_{12})(C_{11} + 2C_{12})}{C_{11} + C_{12}}, \quad \nu = \frac{C_{12}}{C_{11} + C_{12}}, \quad B = \frac{1}{3}(C_{11} + 2C_{12}).$$

A stiff alloy resists stretching (E large); a low ν means it barely thins under load. Because stiffness fixes deflection, vibration and buckling, E and ν are the first properties screened when selecting a structural alloy — and the targets this pipeline predicts and optimizes.

THE BOTTLENECK

Molecular dynamics recovers E and ν cheaply — but only with an accurate interatomic potential. The standard metallic form, the Modified Embedded Atom Method (MEAM) [1, 2], is fitted *by hand*: expert work that does not scale past a few elements.

Objective. Build MEAM potentials *automatically* for any set of elements — no hand fitting, no experimental input.

THE DESIGN SPACE

Compositions are drawn on the 12-element simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn), weighted toward aluminium. Of ~ 7900 random configurations, **7815 survive** the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).

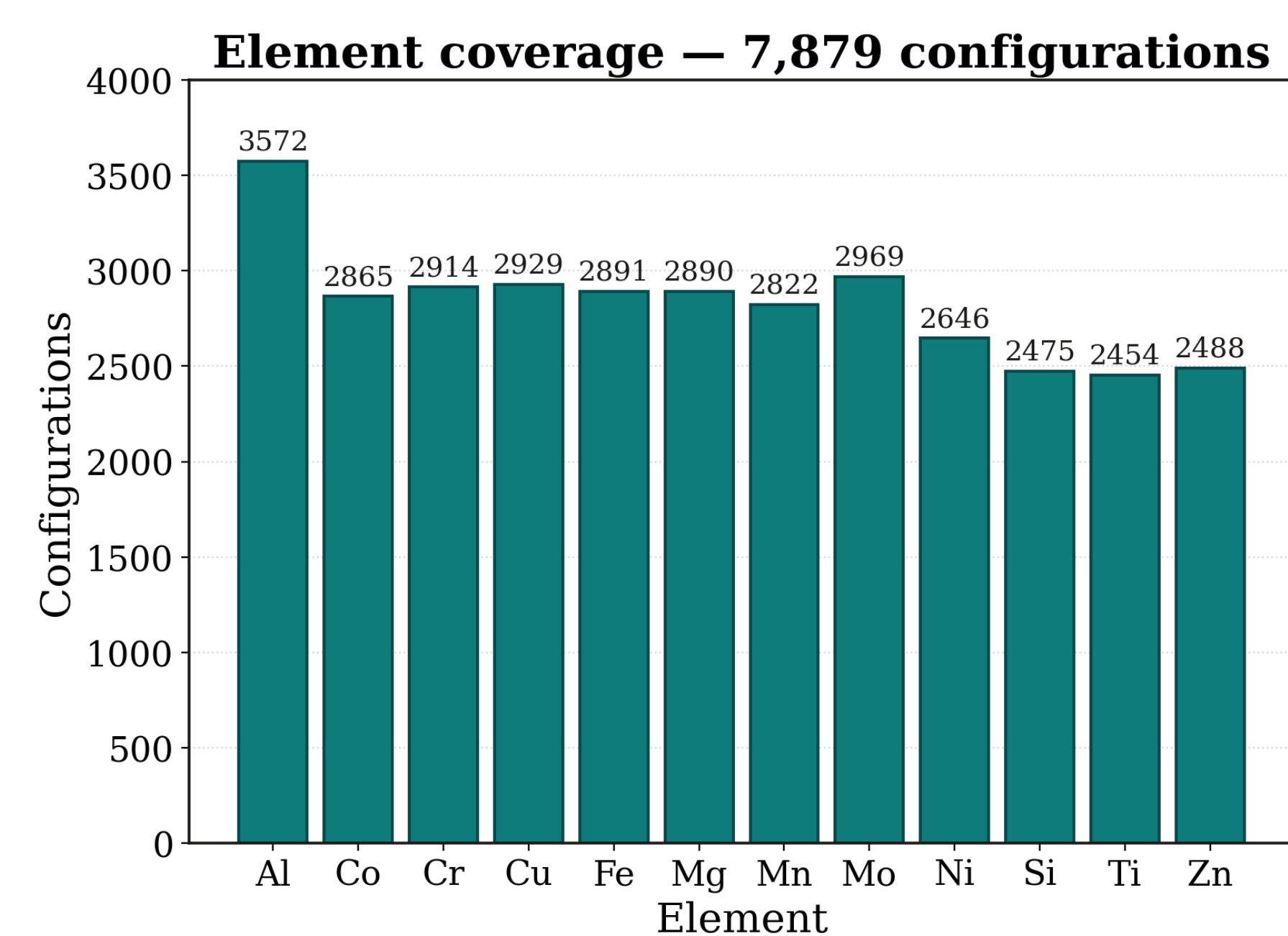


Fig. 1. Element coverage across the retained corpus.

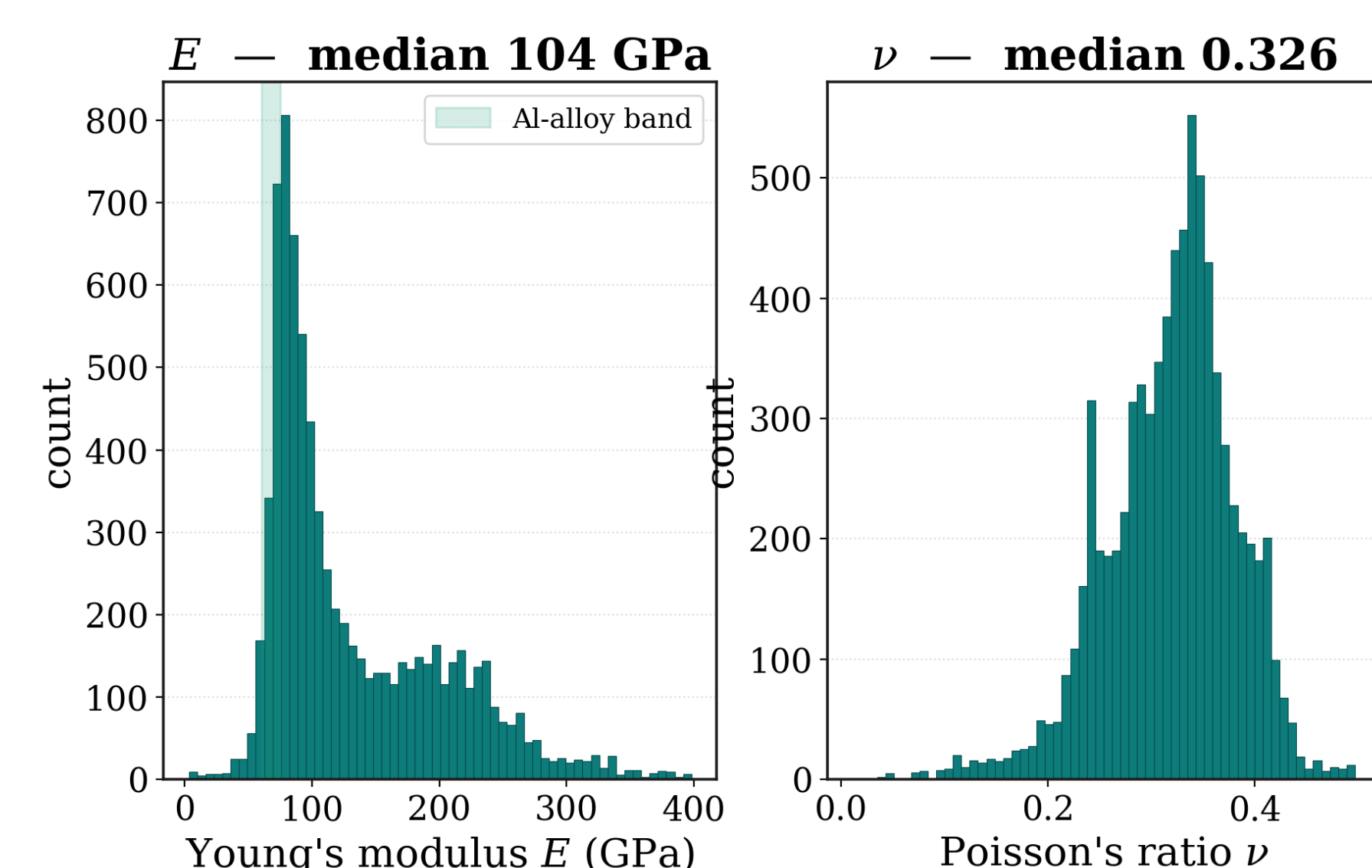
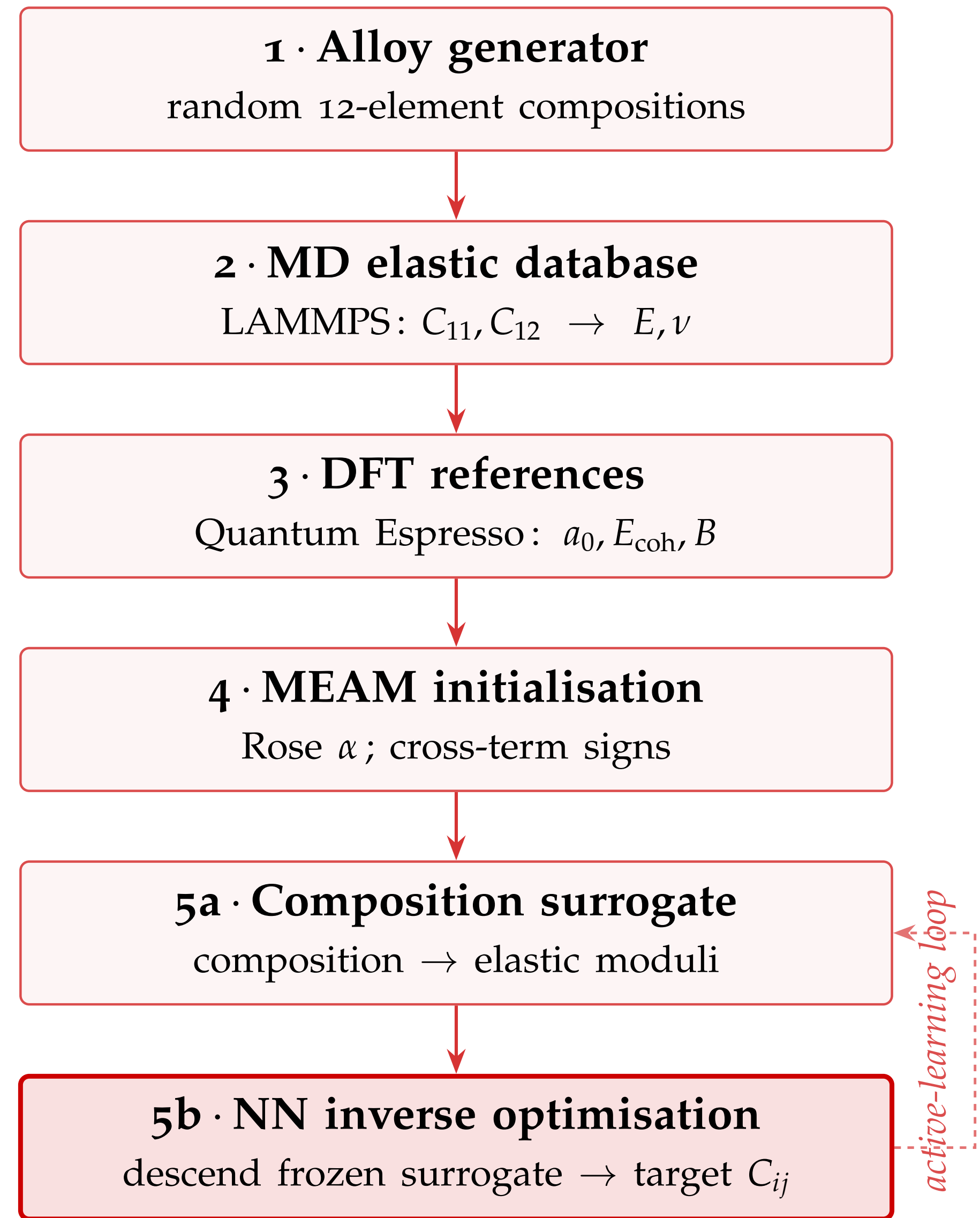


Fig. 2. Property space: E and ν over the 7815-alloy corpus.

2 Methodology

An automated five-stage pipeline turns random compositions into an optimized multi-element potential.



DFT-SEEDED INITIALISATION

Quantum Espresso [4] fixes each element's a_0 , E_{coh} and bulk modulus B , pinning the Rose binding curve [5] and the MEAM decay exponent

$$\alpha = \sqrt{9B\Omega/E_{\text{coh}}}$$

(Ω the atomic volume; $\alpha \approx 4.5$ for Al, Si up to 10.9 for Zn). Sixty-six binary formation energies seed each cross-term's sign.

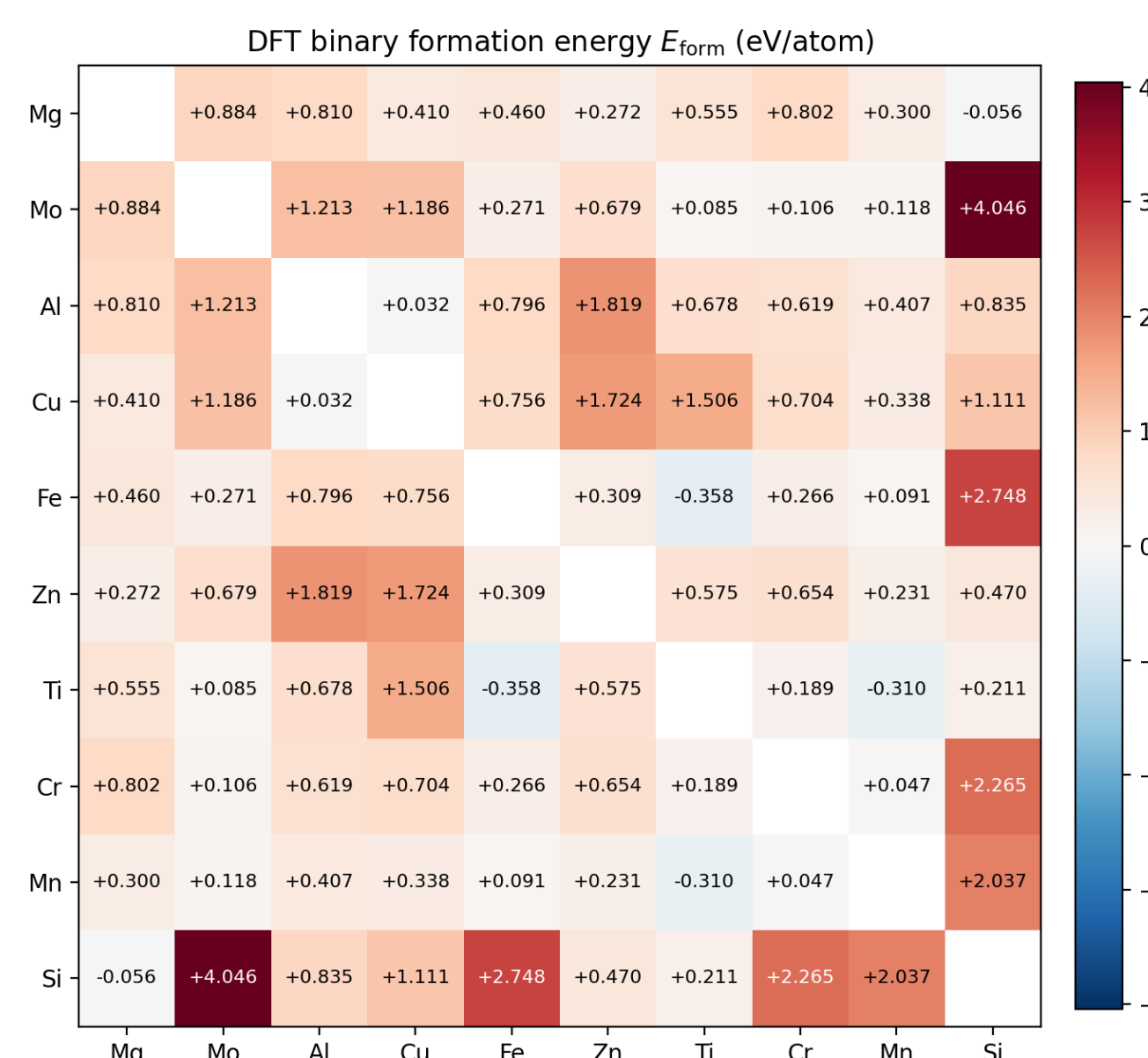


Fig. 3. DFT binary formation energies (eV/atom); negative = exothermic.

NEURAL SURROGATE & INVERSE DESIGN

A fully connected network maps composition to the stiffness constants (E , ν follow analytically). A second network learns (MEAM parameters) $\rightarrow (C_{11}, C_{12})$; gradient descent through the *frozen* surrogate then locates parameters that hit a target.

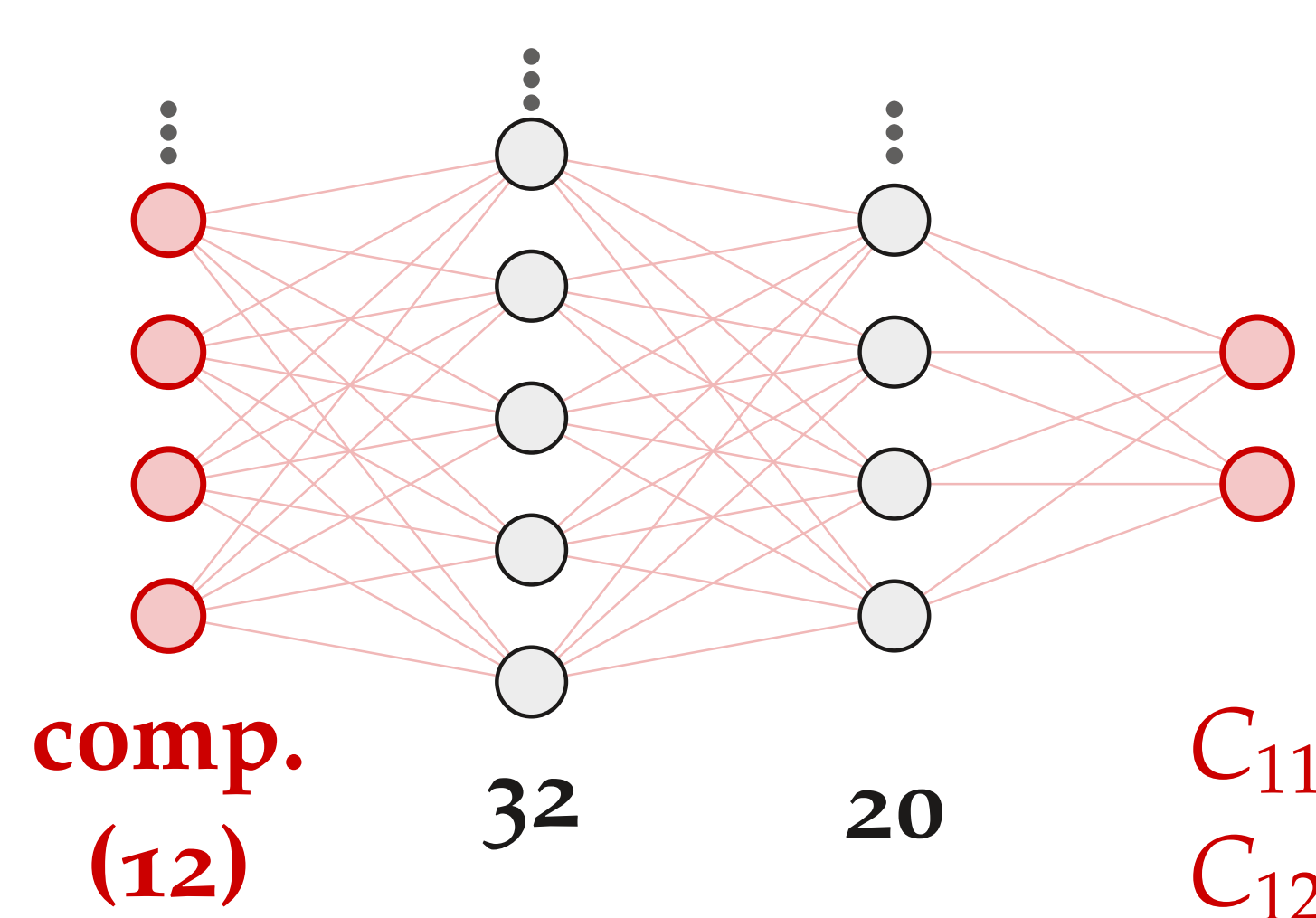


Fig. 4. Surrogate architecture: composition $\rightarrow$ two ReLU hidden layers $\rightarrow (C_{11}, C_{12})$.

3 Results & discussion

ELASTIC DATABASE

A ~ 4 nm LAMMPS [3] supercell per alloy; axial strains ($\delta = 10^{-3}$) give C_{11}, C_{12} and hence E and $\nu = C_{12}/(C_{11} + C_{12})$. Median $E = 103.6$ GPa, $\nu = 0.326$.

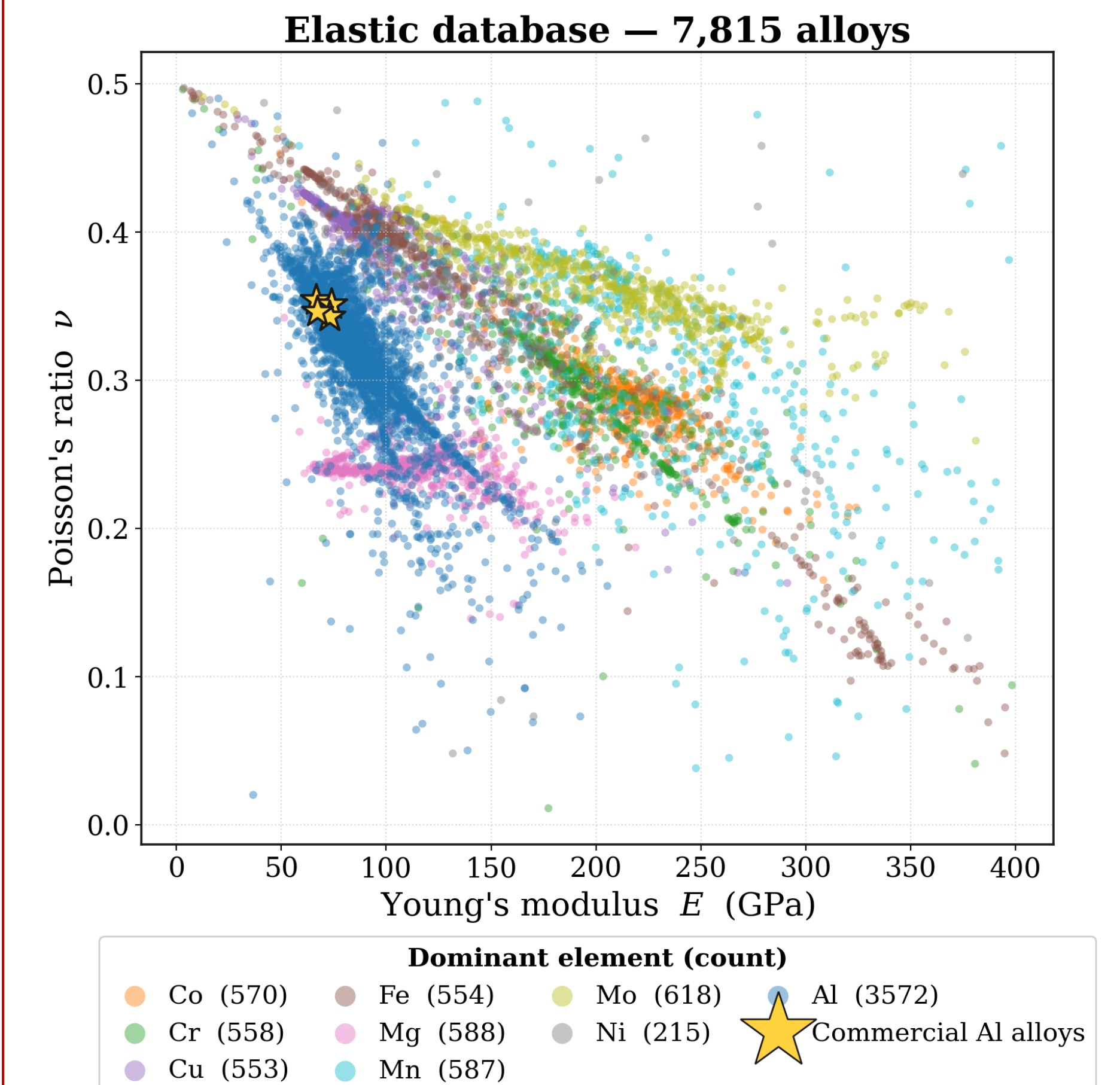


Fig. 5. E - ν for 7815 alloys, coloured by dominant element; commercial Al alloys starred.

COMPOSITION SURROGATE

$R_E^2 = 0.77$, $R_\nu^2 = 0.52$ on held-out validation. For Al₇₀Fe₃₀ the surrogate gives $E = 75.5$ GPa, $\nu = 0.334$ vs published 71–72 GPa, $\nu \approx 0.33$ — within $\sim 5\%$.

OPTIMIZED POTENTIAL

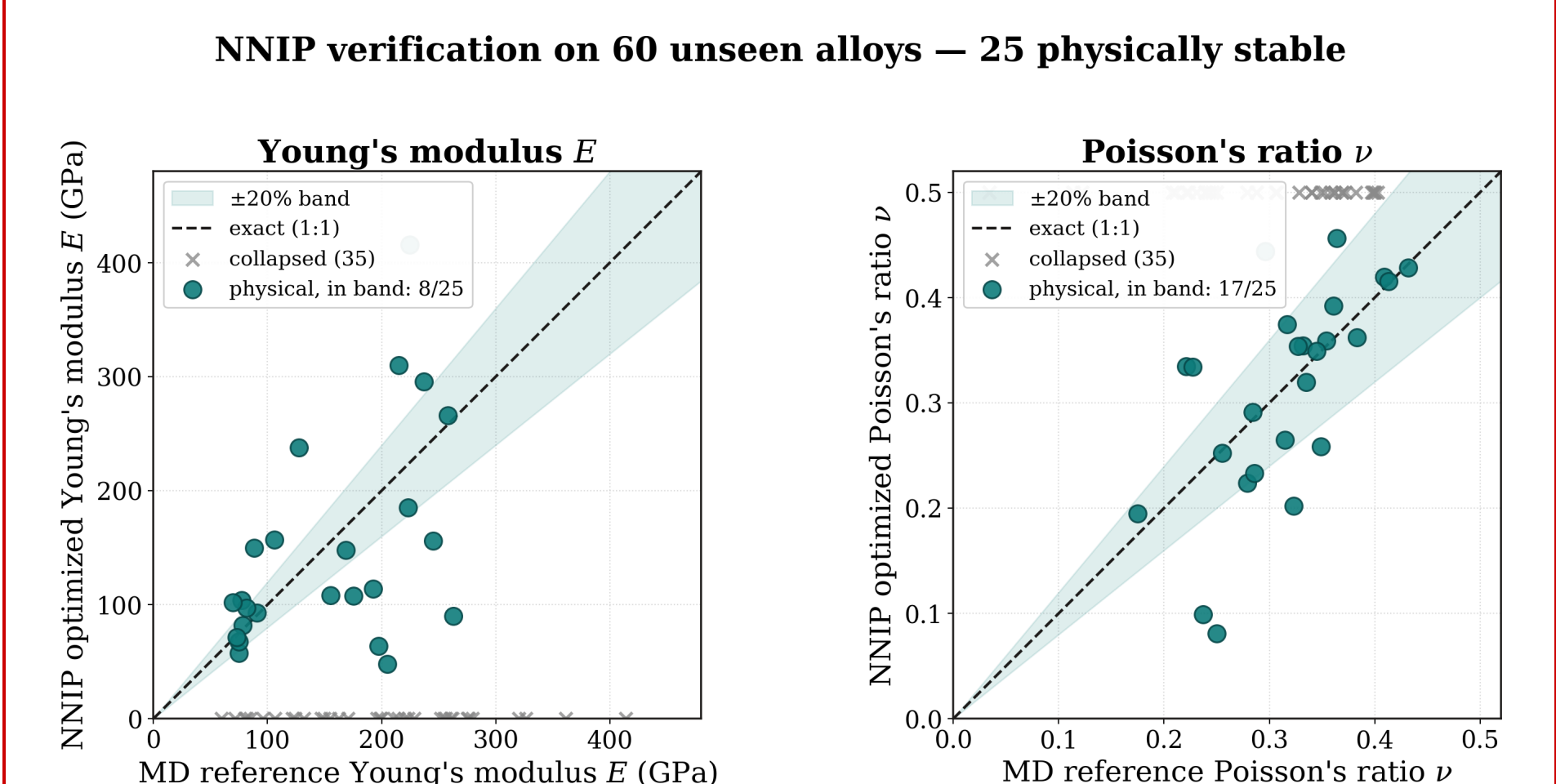


Fig. 6. Optimized potential vs MD on 60 unseen alloys: inside the $\pm 20\%$ band = match, $\times$ = collapsed (unphysical).

Proof of concept. Of 60 unseen alloys, 25 stayed physically stable; 17 matched ν and 8 matched E within $\sim 20\%$.

VALIDATION VS PUBLISHED DATA

Family	N	E MAPE (%)		ν MAPE (%)	
		ML	MEAM	ML	MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Overall	43	10.8	42.2	7.2	7.1

Table 1. Surrogate (ML) and DFT-initialised potential (MEAM) vs published elastic data. Accuracy peaks where the corpus is densest (aluminium); sparse families inherit its bias — a data limit, not a method limit.

CONCLUSIONS & OUTLOOK

- Automated end-to-end 12-element MEAM pipeline — random compositions to an optimized potential, no hand fitting.
- Al-alloy elastic prediction from composition alone, within a few percent of published values.
- DFT-seeded, NN-trained MEAM — first-principles references refit through a neural surrogate.
- Next: more data, longer training, denser DFT references.